

Chubby Dolphin AI

Enterprise Architecture Blueprint

Complete Implementation Guide — All 12 Missing Features

Phase 1 → Phase 2 → Phase 3 | Production-Grade | India-First

Confidential — 2026

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1. Product vision and competitive position

Chubby Dolphin AI is a fully autonomous digital marketing operations platform that replaces the human ad manager entirely. Unlike Ryze AI, Madgicx, Albert, and Smartly — which are built for the US/European market — Chubby Dolphin is India-first: WhatsApp lead automation, vernacular ad copy in 6 languages, INR billing with GST invoicing, and agency white-labelling built in from day one.

Exclusive moat feature	Global competitors	Your advantage
WhatsApp Business API lead automation	Zero of them have it	India #1 channel — instant lead response
Vernacular copy (Hindi/Tamil/Telugu/Kannada/Bengali/Marathi)	Zero of them have it	22 official Indian languages — you serve all
INR billing + GST invoicing + UPI	Zero of them have it	Indian agencies need this for compliance
Brain decision audit trail (plain English)	Zero of them show WHY AI acted	Builds client trust = retention
White-label agency portal with full RBAC	Only partial in Ryze	Agencies resell under their brand

2. Full enterprise architecture (text diagram)

CLIENT LAYER

- Angular 21 SPA (lazy loading, Angular Material, WebSocket real-time)
- Mobile PWA (offline-capable, push notifications)
- White-label agency portal (branded per workspace)

API GATEWAY

- Spring Cloud Gateway — rate limiting, JWT validation, CORS, DDoS shield
- Per-tenant rate limits, request correlation IDs, circuit breaker

CORE MICROSERVICES

- Auth Service — OAuth2, JWT, 2FA (TOTP), RBAC, session management
- Campaign Service — Meta API bridge, campaign CRUD, sync scheduler
- Lead Service — AI scoring, CRM push, webhook ingestion
- Brain Service — autonomous decision engine, safety rules, confidence gating
- Creative Service — ad copy gen, vernacular, A/B, fatigue detection
- Billing Service — Razorpay, wallet, usage metering, GST invoicing
- Report Service — PDF generation, white-label, scheduled delivery
- Notification Service — WhatsApp Business API, push, email alerts

META INTEGRATION LAYER

- OAuth2 Connect — long-lived token, multi-account, encrypted storage
- Campaign Executor — CREATE / PAUSE / RESUME / BUDGET via Graph API v21
- Metrics Sync — ROAS, CTR, CPC, CPL, frequency (hourly snapshots)
- Webhooks — lead form submissions, ad events, pixel events
- Conversions API (CAPI) — server-side event tracking (iOS 14+ fix)
- Audience Builder — custom audiences, lookalikes, Super Lookalikes
- Ad Library Spy — competitor creative tracking via Ad Library API

GOOGLE ADS LAYER (Phase 3)

- Google Ads OAuth2 + campaign sync
- Unified cross-platform brain decisions (Meta + Google together)

WHATSAPP AUTOMATION LAYER

- WhatsApp Business API (Meta Cloud API)
- Lead form trigger → instant WhatsApp message (< 5 seconds)
- Template message library (real estate, education, fintech, healthcare)
- Follow-up sequence scheduler (Day 1 / Day 3 / Day 7)
- Conversation history stored per lead

AUTONOMOUS BRAIN PIPELINE

- Step 1 — Ingest: pull live metrics from Meta API for all active campaigns
- Step 2 — Analyse: send to Mistral 7B with structured JSON prompt

- Step 3 — Safety check: SafetyRulesEngine validates hard limits
- Step 4 — Confidence gate: >0.90 auto-execute | 0.70-0.89 execute+notify
- 0.50-0.69 queue for approval | <0.50 log only
- Step 5 — Execute: Graph API call with full error handling + retry
- Step 6 — Log: BrainDecision + BrainEvent with plain-English summary
- Step 7 — Feedback: outcome stored, decision quality score updated
- Creative Fatigue Sub-pipeline: CTR drop >20% OR frequency >4.0 → alert + rotate

AI INFERENCE LAYER

- Primary: Mistral 7B via Ollama (self-hosted, zero inference cost)
- Fallback: Gemini Pro (cloud, graceful degradation)
- Vernacular: translate/generate in Hindi/Tamil/Telugu/Kannada/Bengali/Marathi
- Embeddings: lead similarity search via FAISS

DATA LAYER

- PostgreSQL 16 — primary store, Row-Level Security per tenant
- Redis 7 — cache, sessions, Pub/Sub, distributed locks (ShedLock)
- RabbitMQ 3.12 — brain decisions queue, webhook event queue
- MinIO (S3) — ad creatives, PDF reports, uploaded assets

INFRA + DEVOPS

- Docker Compose — full local dev stack
- Kubernetes + Helm — production, auto-scaling, HA
- GitHub Actions — CI/CD: test → build → Docker Hub → K8s rolling deploy
- Prometheus + Grafana — metrics, brain decision dashboards, alerts
- Spring Vault / env vars — ALL secrets, never in properties files

3. 12 missing features — priority matrix

#	Feature	Priority	Competitor gap	Est. effort
1	Real Meta API execution (pause/budget/resume)	P1 CRITICAL	All have it — you don't yet	3 days
2	Creative fatigue detection engine	P1 CRITICAL	Ryze/Madgicx/Albert/Smartly all have it	2 days
3	WhatsApp Business API lead responder	P1 YOUR MOAT	ZERO competitors have this	4 days
4	Meta Conversions API (CAPI) server-side tracking	P1 CRITICAL	Madgicx/Albert have it	2 days
5	Vernacular ad copy (6 Indian languages)	P2 YOUR MOAT	ZERO competitors have this	3 days
6	AI audience builder (lookalikes + clusters)	P2	Ryze/Madgicx/Albert have it	4 days
7	Competitor ad spy (Meta Ad Library API)	P2	Madgicx has 25M+ ad library	3 days
8	One-click white-label PDF client report	P2	All competitors have this	2 days
9	CRM push integration (Zoho + HubSpot)	P2	Only Albert has partial	3 days
10	Google Ads API integration	P3	Ryze/Albert/Smartly have it	5 days
11	GST-compliant INR invoicing	P3 YOUR MOAT	ZERO competitors offer this	2 days
12	Advantage+ intelligence layer	P3	No third-party does this well	3 days

4. Implementation phases overview

PHASE 1 — Survival (Weeks 1–3)

Feature 1: Real Meta API execution

Wire MetaAdsService to live Graph API. Implement pause, resume, updateBudget. Test on real ad account. This makes the product real.

Feature 2: Creative fatigue detection

Track CTR/frequency per creative hourly. Fire alert + rotation when CTR drops 20%+ or frequency >4.0. Add CreativeFatigueService + scheduler.

Feature 3: WhatsApp Business API

Meta Cloud API integration. Lead form webhook → immediate WhatsApp message. Template library. Follow-up sequences. This is your moat.

Feature 4: Meta CAPI (server-side tracking)

Implement Conversions API Gateway. Send server-side purchase, lead, and page view events. Fix iOS 14+ attribution loss.

PHASE 2 — Growth (Weeks 4–7)

Feature 5: Vernacular ad copy generation

Extend CreativeAIService to generate in Hindi, Tamil, Telugu, Kannada, Bengali, Marathi. Language selector per campaign. Mistral 7B handles multilingual.

Feature 6: AI audience builder

Meta custom audience + lookalike creation via API. Super Lookalike (blend 20 sources). eRFM audience segmentation. Audience performance tracking.

Feature 7: Competitor ad spy

Meta Ad Library API integration. Tag ads by format, hook type, CTA style. Build searchable creative intelligence dashboard per niche/industry.

Feature 8: One-click PDF client report

Auto-generate white-labeled PDF: ROAS, CTR, CPC, spend, leads, brain decisions. Weekly/monthly scheduler. Agency logo + brand colors per workspace.

Feature 9: CRM push integration

Zoho CRM + HubSpot connectors. Hot leads (score > threshold) auto-pushed with campaign source, score, WhatsApp history. Webhook-based, plug-and-play.

PHASE 3 — Scale (Weeks 8–12)

Feature 10: Google Ads API integration

Google Ads OAuth2, campaign sync, metrics pull, budget/pause/resume execution. Unified brain decisions across Meta + Google simultaneously.

Feature 11: GST-compliant INR invoicing

Auto-generate GST invoices for every Razorpay transaction. GSTIN field per workspace. PDF invoice with HSN code, CGST/SGST/IGST breakdown. Email delivery.

Feature 12: Advantage+ intelligence layer

Brain logic to detect when ASC (Advantage+ Shopping Campaigns) would outperform manual targeting. Auto-suggest or auto-switch. Track ASC vs manual ROAS delta.

Copy this ENTIRE prompt at the start of every Phase 1 coding session. Do not summarise or shorten it.

[illegible]

6. Phase 2 — Master development prompt

Use this prompt after Phase 1 is fully verified. Paste it at the start of every Phase 2 coding session.

[illegible]

7. Phase 3 — Master development prompt

STRICT EXECUTION RULES: No questions. Complete all code fully.

PROJECT: Chubby Dolphin

AI – Phase 3 Implementation

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8. Database schema additions

V3 — meta execution fields

campaigns: meta_campaign_id, last_synced_at | brain_events: meta_error_code, execution_status

V4 — fatigue_alerts

id, campaign_id, creative_id, fatigue_type, baseline_ctr, current_ctr, baseline_frequency, current_frequency, detected_at, status, workspace_id

V5 — whatsapp tables

whatsapp_messages(id,lead_id,workspace_id,to_number,template_name,status,sent_at,reply_text) |
whatsapp_templates(id,workspace_id,name,language_code,body_text,industry_type)

V6 — pixel_configs

id, workspace_id, pixel_id, access_token_encrypted, test_event_code, is_active

V7 — creative language

ad_creatives: language VARCHAR(5), is_vernacular BOOLEAN

V8 — audiences

meta_audiences(id,workspace_id,meta_audience_id,name,subtype,size,source_audience_ids,created_at)

V9 — competitor_ads

competitor_ads(id,workspace_id,keyword,page_name,hook_type,format,offer_type,emotion,ad_text,snapshot_url,fetched_at)

V10 — reports

reports(id,workspace_id,report_type,period_from,period_to,pdf_path,sent_at,created_at)

V11 — crm_config

crm_configs(id,workspace_id,crm_type,access_token_encrypted,refresh_token_encrypted,score_threshold,auto_sync_enabled)

V12 — google_ads

google_ads_connections(id,workspace_id,customer_id,developer_token,refresh_token_encrypted,access_token_encrypted,token_expires_at)

V13-14 — invoices

invoices(id,workspace_id,invoice_number,transaction_id,subtotal,gst_amount,total,gst_type,pdf_path,sent_at) | invoice_configs(id,workspace_id,gstin,legal_name,billing_address,state_code,pan)

V15 — advantage experiments

advantage_experiments(id,campaign_id,switched_at,roas_before,roas_after_14d,roas_after_30d,status,net_roas_delta)

9. External API integration reference

API	Base URL	Auth	Used for
Meta Marketing API v21	https://graph.facebook.com/v21.0	Bearer {longLivedAccessToken}	Campaigns, insights, pause, budget, audiences
Meta Conversions API	https://graph.facebook.com/v21.0/{pixelId}/events	access_token param	Server-side purchase, lead, PageView events
Meta Cloud API (WhatsApp)	https://graph.facebook.com/v21.0/{phoneNumberId}/messages	Bearer {whatsappToken}	Template messages, follow-ups, webhook replies
Meta Ad Library API	https://graph.facebook.com/v21.0/ads_archive	No auth for India public ads	Competitor ad creative research
Ollama (local)	http://localhost:11434/api/generate	No auth — local	Mistral 7B — brain decisions, copy generation
Gemini Pro	https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-pro:generateContent	?key={geminiApiKey}	Cloud fallback when Ollama unavailable
Google Ads API v17	https://googleads.googleapis.com/v17	OAuth2 refresh token flow	Campaigns, budgets, insights, pause/resume
Zoho CRM v3	https://www.zohoapis.in/crm/v3	Zoho-oauthtoken {token}	Push hot leads as CRM contacts
HubSpot CRM v3	https://api.hubapi.com/crm/v3	Bearer {privateAppToken}	Push hot leads as HubSpot contacts
Razorpay API	https://api.razorpay.com/v1	Basic auth key:secret	Payments, subscriptions, webhook events

10. Verification checklist per phase

Phase 1 — must all pass before starting Phase 2

- curl POST /api/campaigns/{id}/pause → 200 OK + BrainEvent in DB with action=PAUSE
- curl POST /api/campaigns/{id}/resume → 200 OK + Campaign status=ACTIVE on Meta
- curl POST /api/campaigns/{id}/budget → Budget updated on real Meta campaign
- Simulate webhook: curl POST /webhooks/meta/leads → Lead created + WhatsApp message sent
- Check frequency > 4.0 scenario → FatigueAlert created, creative rotation triggered
- Verify SHA-256 hashing on CAPI events: no raw PII sent to Meta
- psql: SELECT * FROM fatigue_alerts; — should have records
- psql: SELECT * FROM whatsapp_messages; — should have records after lead test
- mvn test — ALL tests pass with zero failures
- grep -r 'TODO\|return null\|UnsupportedOperation' src/main/java/ — zero results

Phase 2 — must all pass before starting Phase 3

- Generate Hindi ad copy: POST /api/creative/generate with language=hi → 5 Hindi variants returned
- Generate Tamil ad copy: language=ta → 5 Tamil variants in Tamil script
- Create lookalike audience via API → audience visible in Meta Business Suite
- Competitor spy search: GET /api/spy/search?keyword=real+estate → tagged competitor ads returned
- Generate PDF report: GET /api/reports/generate?workspaceId=X → PDF downloaded with real data
- Lead with score > 70 → auto-pushed to Zoho CRM → verify in Zoho dashboard
- Check MinIO bucket: reports/ folder has PDF files
- Scheduled report runs on Monday 8 AM — check logs for confirmation

Phase 3 — final enterprise readiness

- Connect Google Ads account → campaigns sync to DB
- Brain cycle runs for both Meta + Google campaigns
- Generate GST invoice: POST /api/billing/invoice/{txnId} → PDF with CGST/SGST breakdown
- Verify invoice has correct SAC code 998314, GSTIN, sequential invoice number
- Advantage+ analysis runs: check advantage_experiments table has records
- All 15 Flyway migrations applied: SELECT version FROM flyway_schema_history ORDER BY installed_rank;
- Load test: 100 concurrent brain cycle requests → no deadlocks, no OOM
- Security scan: no secrets in any .properties or .java file